Tether-Aware Coupled Orienteering of a Catamaran Base and Its Tethered ROV for Time- and Energy-Bounded Seabed Ore Sampling

Yuhang Bao[1,†] and Jiaqing Wang[1,†]

***Abstract*—Seabed ore sampling by a tethered remotely operated vehicle that works from a mobile catamaran mother ship must fit a time budget and an energy budget drawn from one store on the catamaran. However, the tether couples the two vehicles through a reach that depends on the depth of the vehicle and a drag load that grows with the motion of the catamaran. Therefore, we propose the Tether-Aware Coupled Orienteering planner to plan the motion of the catamaran together with the sampling visits. The planner turns the tether limit into one reach disc per visit and charges the load of a moving tether to the energy of each leg. A convex program places the catamaran waypoints and sets the leg durations of every candidate order of visits. An order search keeps the plan with the largest value weighted by the probability that each visit is completed within both budgets. The planner replans after every sample. We fitted the cost model to simulator legs and to seven field dives of a work-class vehicle, whose 78 sampling stations lasted 581 s on average. In two hardware-in-the-loop scenarios with PX4 on a flight controller, the planner kept 6.2% and 17.0% more sample value than a fixed catamaran. On the field dives with budgets cut to 70%, the planner kept 47.1 samples against 44.0 for a fixed catamaran and 45 for the crew of the dives. These results show that planning the catamaran and the tethered vehicle as one coupled problem raises the value of a budgeted sampling dive.**

***Index Terms*—Catamaran, energy budget, hardware-in-the-loop, orienteering, remotely operated vehicle, seabed sampling, tether.**

I. Introduction

Cobalt-rich ferromanganese crusts and polymetallic nodules on the deep seabed hold cobalt, nickel, copper and rare earth elements at grades comparable to land deposits [1]–[3]. Their exploitation is still at the exploration stage. Many target areas are vulnerable ecosystems whose disturbance lasts for decades [4], [5]. Before any collection, a resource estimate rests on physical samples taken at selected points. On one guyot of the western Pacific, a work-class Remotely Operated Vehicle (ROV) measured crust thicknesses from 7 to 15 cm that varied with the seabed slope [6]. The choice of sampling points thus decides the value of a dive. Every sample costs mission time and energy. Automating the manipulator work of such vehicles is still a research topic [7], [8]. In the seven sampling dives of a work-class ROV that we analyzed, a landed sampling station lasted 581 s on average with a standard deviation of 504 s. During each bottom phase the support ship moved 1.5 to 6.6 km to keep the umbilical within reach of the vehicle. Because the surface base follows the vehicle, its plan cannot be made apart from the plan of the ROV. Fig. 1 shows this coupled setting.

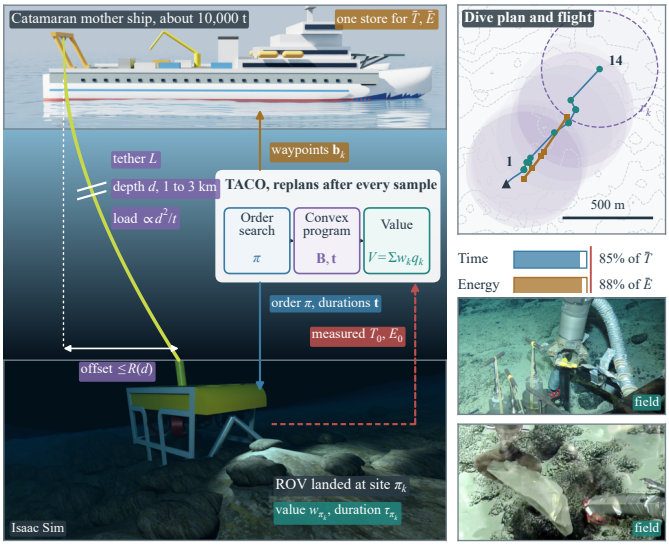


Fig. 1. Tether-Aware Coupled Orienteering (TACO). A catamaran mother ship of about 10,000 t holds station while its tethered work-class ROV samples a seabed site. TACO places the catamaran waypoints inside the reach discs of the visits and sets the leg durations. It weights each visit by its probability of completion within the time budget $\bar{T}$ and the energy budget $\bar{E}$. After every sample it replans from the measured time and energy. The section joins two Isaac Sim renderings of one instant of a closed-loop flight of TACO on a field dive. The water column between them is not to scale. The map shows the plan and the flight of that dive with the reach discs of the visits. Filled bars give the flown shares of both budgets. Outlines give the planned shares. The photograph and the video frame come from field dives of a work-class ROV.

Unmanned surface vehicles already map the seabed by coverage planning [9]. Previous studies have routed a drone from a mothership [10], an aerial vehicle from a ground vehicle [11] and underwater vehicles from a surface vehicle [12], [13]. Eskandari *et al.* [12] planned a surface vehicle and its underwater vehicles jointly under an acoustic range constraint by sequential convex programming. Tethered robots have been planned around fixed anchors [14]–[16]. The umbilical between a surface vehicle and an ROV has been studied as a control problem [17]. For the choice of visits, Carpin [18] bounded the probability that a stochastic orienteering plan exceeds its budget. However, these methods have not solved the coupling that a tether creates. The reach of the ROV depends on its depth and on the position of the base. Moving

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the base sweeps the tether through the water and costs the ROV energy. The base motion therefore enters the energy budget of the dive.

Motivated by this, we propose the Tether-Aware Coupled Orienteering (TACO) planner for a catamaran mother ship that deploys a tethered ROV to sample the seabed. The time budget and the energy budget of the dive are drawn from one store on the catamaran. To our knowledge, no published method plans the motion of the anchor of a tethered underwater vehicle together with its sampling visits. Published sampling planners also do not weight their visits by sampling durations measured in field operations. TACO turns the tether constraint into one reach disc per visit and charges the load of a moving tether to the leg energy. A convex program then places the catamaran waypoints and sets the leg durations for every candidate order of visits. An order search keeps the plan with the largest value weighted by the probability that each visit is completed within both budgets. TACO plans again after every sample. We fit its leg energy model to calibration missions in NezhaSim2, an in-house simulator that is not yet open source. The model of a work-class vehicle comes from the logs of field dives. Besides, we fly the ROV with PX4 on a Pixhawk 6C flight controller in hardware-in-the-loop. The hardware-in-the-loop scenarios test the closed loop at a reduced scale. The field scenario tests TACO at the scale of a work-class ROV and its mother ship.

Our contributions are summarized as follows.

- We formulate budgeted seabed sampling by a catamaran base and a tethered ROV as an orienteering problem with reach discs and synchronized legs. We show that reach discs at the leg ends bound the tether offset along each leg. The leg energy carries the tether load of a moving base.
- We propose TACO, which places the catamaran waypoints and sets the leg durations by a convex program. TACO selects the visits by a value weighted with their completion probabilities and replans after every sample.
- We identify the cost model from simulator legs and from seven dives of a work-class ROV, including the duration distribution of 78 field sampling stations. We evaluate TACO in two PX4 hardware-in-the-loop scenarios and one field-data scenario against a fixed-base planner and a cluster-first planner. In hardware in the loop TACO kept 6.2% and 17.0% more value than the fixed-base planner. On the field dives at 70% of their budgets it kept 7.0% more samples. On 29 of 40 small instances it matched or exceeded an exhaustive search over visit orders.[1]

II. Related Works and Existing Gaps

A. Mobile Bases of Range-Limited Vehicles

Poikonen and Golden [10] defined the mothership and drone routing problem in the open sea. In their problem a drone with a limited range leaves a mothership, visits targets and lands on it again. They chose the launch and landing points by a second-order cone program inside a branch-and-bound over target sequences. Mondal *et al.* [11] used a ground vehicle as a mobile recharging base of an energy-limited aerial vehicle. They compared a learned policy with a bilevel planner that routes the base first and the aerial vehicle second. Eskandari *et al.* [12] planned a surface vehicle and a team of underwater vehicles jointly by model predictive control. Every underwater vehicle had to stay within the acoustic range of the surface vehicle. They solved the problem by sequential convex programming. Cao *et al.* [13] combined aerial, surface and underwater vehicles for underwater target capture under communication and energy constraints. Sarda and Dhanak [19] launched and recovered an underwater vehicle from a station-keeping catamaran.

In all of these studies the link between the base and the vehicle is a range, a communication edge or a recharge event. The link itself costs no energy and does not depend on depth. None of them charges the energy of moving a physical connection to the plan.

B. Tether-Constrained Robot Planning

Kim *et al.* [14] planned paths of a planar robot tied to a fixed base by a cable among obstacles. Shnaps and Rimon [15] covered unknown planar environments with a robot tied to a fixed point by a cable of fixed length. Petit and Lussier Desbiens [16] split the exploration of a tethered aerial robot into a global frontier tour and a local planner that keeps the unwound tether within its length. For underwater vehicles, Laranjeira *et al.* [20] estimated and controlled the catenary shape of a tether between two vehicles. Tortorici *et al.* [17] derived coordinated winch, surface vehicle and underwater vehicle control from simulated cable loads of a shallow-water system. Driscoll *et al.* [21] validated a lumped-mass model of a deep-sea vehicle tether against at-sea data. Von Benzon *et al.* [22] coupled a lumped-mass tether with normal and tangential drag to an identified model of a small eight-thruster vehicle.

These studies anchor the tether at a fixed point, or they treat the cable as a problem of low-level control. None of them plans the motion of the tether anchor together with the visits of the tethered vehicle.

C. Budgeted Orienteering Under Uncertainty

Golden *et al.* [23] introduced the orienteering problem, which selects and orders visits to maximize the collected score under a travel budget. Faigl *et al.* [24] solved a variant in which the reward of a location is collected anywhere inside a disc around it. Coutinho *et al.* [25] bounded the close-enough traveling salesman problem by second-order cone programs that place the visit points inside such regions. Campbell *et al.* [26] defined the orienteering problem with stochastic travel and service times. Carpin [18] bounded the probability of exceeding the budget of a stochastic orienteering problem and alternated planning with execution. Blackmore *et al.* [27] replaced chance constraints of linear Gaussian vehicles with deterministic bounds. At sea, Tsiogkas *et al.* [28] planned energy- and time-constrained sensing missions of an underwater vehicle as a correlated orienteering problem. Eimeni and Khosravi [29] set the speed of an underwater vehicle online

[1]The project page with the video of this work is at https://panzerjagerwang.github.io/TACO-Tethered-ROV/.

from its state of charge. Zhao *et al.* [30] scheduled time-sensitive rescue tasks of surface vehicles with an accumulated reward. Fan *et al.* [31] allocated multi-sortie inspection of an offshore wind farm with battery exchanges. Chirala *et al.* [32] routed teams of leader and follower vehicles by a construction heuristic and variable neighborhood search.

These planners give each budget to the vehicle that spends it. The service-time distributions are assumed rather than measured. None of them draws a time budget and an energy budget from one store shared by two coupled vehicles.

III. Problem Formulation and Preliminaries

A. Budgeted Sampling With a Tethered Vehicle

A catamaran mother ship starts at the surface point $\mathbf{b}_0 \in \mathbb{R}^2$ with the tethered ROV on deck. A prior survey gives N candidate sites $\mathcal{S} = \{1, \dots, N\}$. Site i lies at the seabed point $\mathbf{s}_i \in \mathbb{R}^3$ with horizontal part $\bar{\mathbf{s}}_i$. Its value is $w_i > 0$. Taking a sample at site i lasts the random duration τ_i with mean μ_i and standard deviation σ_i. A plan is the triple $\mathcal{P} = (\boldsymbol{\pi}, \mathbf{B}, \mathbf{t})$, where $\boldsymbol{\pi} = (\pi_1, \dots, \pi_K)$ is an ordered subset of $\mathcal{S}$, $\mathbf{B} = (\mathbf{b}_1, \dots, \mathbf{b}_K)$ holds the catamaran waypoints, and $\mathbf{t} = (t_1, \dots, t_K)$ holds the leg durations. Before the first leg the catamaran may carry the ROV on deck to its first waypoint, where the ROV is deployed. On leg k the ROV moves from its previous point to site π_k while the catamaran moves from $\mathbf{b}_{k-1}$ to $\mathbf{b}_k$. The catamaran then holds $\mathbf{b}_k$ while the ROV samples. After the last sample the ROV ascends to the catamaran.

The tether of length L hangs from a tow point at depth d_c. The horizontal offset of the ROV from the catamaran must not exceed the reach at the ROV depth d

$$R(d) = \eta\sqrt{L^2 - (d - d_c)^2} - \delta, \tag{1}$$

where $\eta < 1$ denotes the usable share of the geometric reach, and δ denotes an allowance for tracking error. For an ROV whose umbilical is far longer than the water depth, R is a constant set by the operating envelope of the support vessel. Leg k has the horizontal ROV path h_k, the vertical excursion u_k of a terrain-following path and the catamaran move $d_k = \|\mathbf{b}_k - \mathbf{b}_{k-1}\|$. Its duration is bounded below by

$$t_k \geq t_k^{\mathrm{r}} = \max\left(\frac{h_k}{v}, \frac{u_k}{v_{\mathrm{z}}}\right) + t_{\mathrm{s}}, \qquad t_k \geq \frac{d_k}{\bar{v}}, \tag{2}$$

where v and v_{z} denote the horizontal and vertical speeds of the ROV, t_{s} denotes the time to settle at a site, and $\bar{v}$ denotes the top speed of the catamaran. The mission time and the energy drawn from the store on the catamaran are

$$T = \sum_{k=1}^{K} (t_k + \tau_{\pi_k}) + t_{\mathrm{a}}, \tag{3}$$

$$E = \sum_{k=1}^{K} (e(t_k, h_k, u_k, d_k) + P_{\mathrm{s}}\tau_{\pi_k}) + e_{\mathrm{a}}, \tag{4}$$

where $e(\cdot)$ denotes the energy of a leg, P_{s} denotes the power drawn while the ROV samples, and t_{a} and e_{a} denote the duration and energy of the ascent. The ROV, its sampling tool and the catamaran all draw on the same store. One energy budget $\bar{E}$ therefore covers both vehicles. For a small electric catamaran $\bar{E}$ is the charge of its battery. For a work-class ROV on a large mother ship it is the hydraulic energy allotted to the dive.

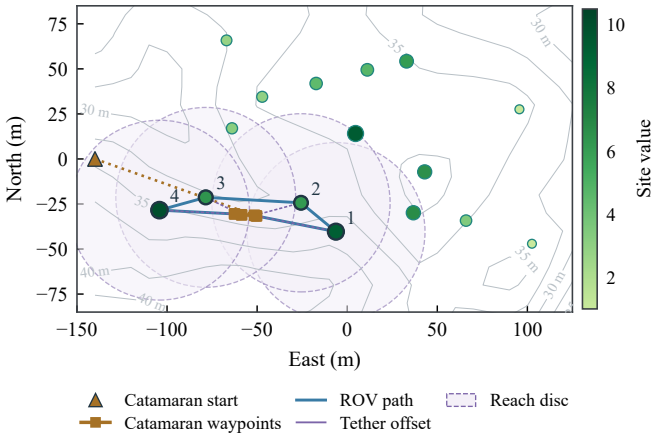


Fig. 2. Planning problem on an instance of Scenario 1. Depth contours come from the relief of a seamount patch. Circles are sites sized and colored by value. Outlined circles are the visits planned by TACO in their order. Dashed discs are the reach discs of the visits. Squares are the catamaran waypoints. The dotted line is the move with the ROV on deck.

Visit k counts only when the ROV completes it and can still return within the time budget $\bar{T}$ and the energy budget $\bar{E}$. The time and energy used when visit k ends are T_k and E_k. The ascent from site π_k takes the time $t_{\mathrm{a},k}$ and the energy $e_{\mathrm{a},k}$. The planner solves

$$\max_{\mathcal{P}} \mathbb{E}\left[\sum_{k=1}^{K} w_{\pi_k} \mathbb{1}\left(T_k + t_{\mathrm{a},k} \leq \bar{T},\ E_k + e_{\mathrm{a},k} \leq \bar{E}\right)\right] \tag{5}$$
$$\text{s.t. } \|\bar{\mathbf{p}}(\theta) - \mathbf{b}(\theta)\| \leq R\big(d(\theta)\big) \quad \forall\theta,$$

where $\mathbb{1}(\cdot)$ denotes the indicator function. The symbols $\bar{\mathbf{p}}(\theta)$, $d(\theta)$ and $\mathbf{b}(\theta)$ denote the horizontal ROV position, the ROV depth and the catamaran position at time θ. The problem is solved online. After every sample the planner knows the time and energy already used, the positions of both vehicles and the sites still open. A mission that exceeds a budget scores zero, because an exhausted store or a missed recovery window loses the samples in practice.

B. Leg Coupling and Notation

Both vehicles move on leg k with one common progress $\lambda \in [0, 1]$. At progress λ the ROV is at $(1-\lambda)\bar{\mathbf{s}}_{\pi_{k-1}} + \lambda\bar{\mathbf{s}}_{\pi_k}$ and the catamaran at $(1-\lambda)\mathbf{b}_{k-1} + \lambda\mathbf{b}_k$. The smallest reach (1) over the depths of the terrain-following path of leg k is ρ_k.

Proposition 1: If $\|\bar{\mathbf{s}}_{\pi_{k-1}} - \mathbf{b}_{k-1}\| \leq \rho_k$ and $\|\bar{\mathbf{s}}_{\pi_k} - \mathbf{b}_k\| \leq \rho_k$, the tether offset stays within ρ_k along the whole leg.

Proof: The offset at progress λ is $(1-\lambda)(\bar{\mathbf{s}}_{\pi_{k-1}} - \mathbf{b}_{k-1}) + \lambda(\bar{\mathbf{s}}_{\pi_k} - \mathbf{b}_k)$. Its norm is at most $(1-\lambda)\rho_k + \lambda\rho_k = \rho_k$ by the triangle inequality. □

Proposition 1 turns the continuous constraint of (5) into one disc per visit,

$$\|\mathbf{b}_k - \bar{\mathbf{s}}_{\pi_k}\| \leq r_k = \min\left(R(d_{\pi_k}), \rho_k, \rho_{k+1}\right), \tag{6}$$

TABLE I
SYMBOLS

Symbol	Meaning
$\mathbf{s}_i$, w_i	seabed point and value of site i
τ_i, μ_i, σ_i	sampling duration, its mean and standard deviation
$\boldsymbol{\pi}$, K	visit order and number of visits
$\mathbf{b}_k$, t_k	catamaran waypoint and duration of leg k
h_k, u_k, d_k	ROV horizontal path, vertical excursion, catamaran move
L, $R(d)$, r_k	tether length, reach at depth d, reach disc of visit k
v, v_z, $\bar{v}$	ROV horizontal and vertical speeds, catamaran top speed
t_s	settling time at a site
$e(\cdot)$, P_s	leg energy, power while sampling
$\bar{T}$, $\bar{E}$	time budget, energy budget
q_k, ϵ	completion probability of visit k, quantile level of U
γ	allowance of the return guard on the ascent
c_t, c_e	variation coefficients of leg duration and leg energy
U	budget utilization of a plan

where d_{π_k} denotes the sampling depth at site π_k. The catamaran is therefore free to stand anywhere in a disc around each site, as in the orienteering problem with neighborhoods [24]. Fig. 2 shows these discs on an instance of Scenario 1. Unlike that problem, the position chosen in the disc changes the energy of both vehicles.

Table I lists the symbols. We assume that sampling durations are independent across sites. Leg durations and leg energies deviate from their model values by independent factors with the coefficients of variation c_t and c_e. The two PX4 scenarios run at a reduced scale with the hull, thrusters and drag of an eight-thruster BlueROV2 Heavy identified in a test tank [22]. Their catamaran is scaled from the surface vessel of our simulator. The field scenario uses the logs of seven dives of a work-class ROV on cobalt-rich crust seamounts of the western Pacific.

IV. TETHER-AWARE COUPLED ORIENTEERING

Two-echelon planners fix the stations of the base before they route the vehicle [33], [34]. Mothership planners give the link between the two vehicles no energy cost [10], [11]. With a tether, the catamaran position decides which sites the ROV can reach and how much energy the tether costs on the way. Therefore, we propose the Tether-Aware Coupled Orienteering (TACO) planner. It places the catamaran inside the reach discs for every candidate order and charges the tether load to the plan. Fig. 3 shows its three parts. Section IV-A gives the coupled leg model. Section IV-B gives the risk-weighted value and the convex waypoint program. Section IV-C gives the order search and its receding-horizon execution.

A. Coupled Leg Model

We model the energy of a leg of duration t as

$$e(t,h,u,d) = \eta_\mathrm{t}^{-1}\Big(a_0 t + a_1\frac{h^3}{t^2} + a_2 u + a_3\frac{d^2}{t} + a_4\frac{d^3}{t^2} + P_\mathrm{h} t\Big) + b_0 t + b_1 d + b_2\frac{d^3}{t^2} + P_\mathrm{c} t, \tag{7}$$

where h denotes the horizontal ROV path, u denotes the vertical excursion, and d denotes the catamaran move. The coefficients a_0 to a_4 belong to the ROV thrust and b_0 to b_2 to the catamaran thrust. The hotel loads of the two vehicles are P_h and P_c. The efficiency of the power conversion and the tether is η_t. The term $a_1 h^3/t^2$ follows from a thrust power that grows with the cube of the cruise speed h/t. The term $a_3 d^2/t$ is the tether load. A catamaran that moves during a leg sweeps the tether through the water. The ROV must then push against the drag of the swept cable. This load grows with the catamaran speed d/t and lasts for the leg duration. A sample of duration τ costs $P_\mathrm{s}\tau$ with $P_\mathrm{s} = (P_\mathrm{hold} + P_\mathrm{h} + P_\mathrm{tool})/\eta_\mathrm{t} + P_\mathrm{c,hold} + P_\mathrm{c}$. Here P_hold and $P_\mathrm{c,hold}$ are the hold powers of the ROV and the catamaran. The power of the sampling tool is P_tool.

We fit the coefficients of (7) by non-negative least squares to the legs of calibration missions in the simulator. The hold powers come from the sampling and planning intervals of the same missions. For the work-class vehicle of the field scenario, the same terms come from the operating modes of the dive logs in Section V-E.

B. Risk-Weighted Value and Convex Waypoints

For a fixed order $\boldsymbol{\pi}$ the waypoints $\mathbf{B}$ and the durations $\mathbf{t}$ remain free. At the end of visit k the time used has the mean $m_{T,k} = T_0 + \sum_{l\le k}(t_l + \mu_{\pi_l})$ and the variance $s^2_{T,k} = \sum_{l\le k}\big((c_t t_l)^2 + \sigma^2_{\pi_l}\big)$, where T_0 denotes the time already used. The energy used has the mean $m_{E,k}$ and the variance $s^2_{E,k}$. They come in the same way from E_0, c_e and P_s. Both are sums of independent terms. We therefore take them as normal. Section V-G compares the resulting probabilities with Monte Carlo draws. Under the return guard of Section IV-C, the probability that visit k counts in (5) is then

$$q_k = \Phi\left(\min\left(\frac{\bar{T} - m_{T,k} - \gamma t_{\mathrm{a},k}}{\tilde{s}_{T,k}},\ \frac{\bar{E} - m_{E,k} - \gamma e_{\mathrm{a},k}}{\tilde{s}_{E,k}}\right)\right), \tag{8}$$

where Φ denotes the standard normal distribution function. The constant $\gamma = 1.2$ is the allowance of the guard on the ascent. The terms $\tilde{s}_{T,k}$ and $\tilde{s}_{E,k}$ add the spread of the ascent to $s_{T,k}$ and $s_{E,k}$. The objective of (5) becomes the risk-weighted value $V = \sum_k w_{\pi_k} q_k$. A visit with $q_k < 0.5$ is more likely cut by the return guard than completed. TACO therefore ends the plan before such a visit.

TACO places the waypoints by a convex surrogate of V. The surrogate is the budget utilization of the whole plan

$$U(\mathbf{B},\mathbf{t}) = \max\left(\frac{\Delta_T + z_\epsilon \tilde{s}_{T,K}}{\bar{T} - T_0},\ \frac{\Delta_E + z_\epsilon \tilde{s}_{E,K}}{\bar{E} - E_0}\right), \tag{9}$$

where $\Delta_T = m_{T,K} - T_0 + t_\mathrm{a}$ and $\Delta_E = m_{E,K} - E_0 + e_\mathrm{a}$ denote the mean time and energy still to be spent with the ascent. The value z_ϵ is the $1-\epsilon$ quantile of the standard normal distribution. A lower U leaves more of both budgets after the last visit. The waypoints and durations of an order come from

$$\min_{\mathbf{B},\mathbf{t}}\ U(\mathbf{B},\mathbf{t}) \quad \text{s.t. } \|\mathbf{b}_k - \bar{\mathbf{s}}_{\pi_k}\| \le r_k,\quad t_k \ge t^\mathrm{r}_k,\quad t_k \ge \frac{\|\mathbf{b}_k - \mathbf{b}_{k-1}\|}{\bar{v}}, \tag{10}$$

for $k = 1,\dots,K$.

Proposition 2: Program (10) is convex.

Proof: The move $d_k = \|\mathbf{b}_k - \mathbf{b}_{k-1}\|$ is convex in $\mathbf{B}$. The functions t, h^3/t^2, d^2/t and d^3/t^2 are convex for $t > 0$. The

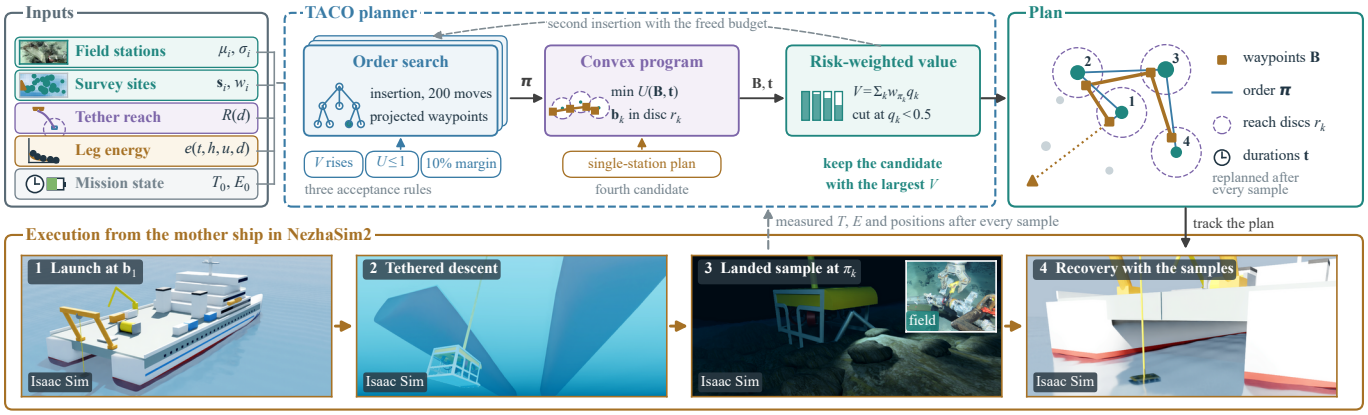


Fig. 3. Tether-Aware Coupled Orienteering (TACO). The order search runs under three acceptance rules with projected waypoints. A single-station plan is a fourth candidate. The convex program places the catamaran waypoints and sets the leg durations of an order. The risk-weighted value selects the plan. The executor tracks the plan and returns the measured state after every sample. The renderings show a catamaran mother ship of about 10,000 t with the work-class ROV of Scenario 3. The ROV stands at logged states of a NezhaSim2 run. The video frame and the photograph come from sampling stations of the field dives.

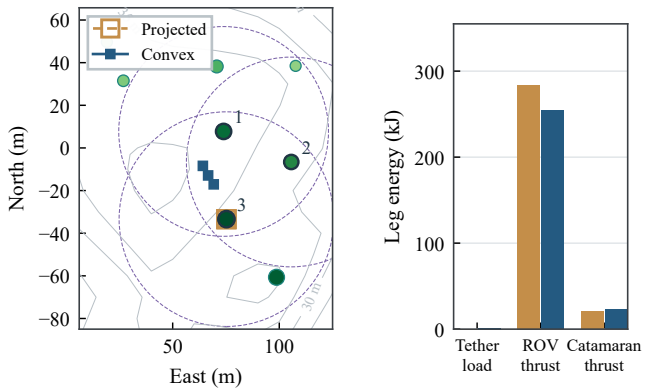


Fig. 4. Projected and convex catamaran waypoints for the visit order of TACO on an instance of Scenario 2, with the leg energy terms of both. Dashed circles are the reach discs. Numbers give the visit order. The projected waypoints coincide above the last site.

last two are nondecreasing in $d \geq 0$. Each term of (7) is therefore convex in $(\mathbf{B}, \mathbf{t})$. The means are sums of convex terms. The standard deviations are Euclidean norms of vectors of nonnegative convex terms and are therefore convex. The maximum of convex functions is convex. Every constraint set is convex. □

A local minimum of (10) is therefore a global one. The cone programs of mothership routing also place the launch points inside regions [10]. In (10) the leg durations are variables too. The base motion further enters the energy of the tethered vehicle. We solve the program by sequential least-squares quadratic programming from the projected waypoints of Section IV-C. TACO keeps the solution only when it does not lower V. A longer leg costs time but lowers the tether load $a_3 d^2/t$. The program therefore decides how fast the catamaran moves on each leg. Fig. 4 shows the trade on an instance of Scenario 2. The convex waypoints shorten the descent to the first site. This lowers the ROV thrust energy from 284.0 to 254.1 kJ for a tether load of 1.5 kJ. The value V rises from 19.9 to 21.5.

C. Order Search and Receding-Horizon Execution

The order search evaluates many candidate orders. The cost of one evaluation grows linearly with the number of visits of the order. For each candidate it uses projected waypoints. The waypoint $\mathbf{b}_k$ is the projection of $\mathbf{b}_{k-1}$ onto the disc of visit k. Its duration t_k is the larger bound of (2). While the ROV is still on deck, the catamaran carries it to the deployment point at no tether cost. A backward pass that projects the last site through the discs to the first one then gives the deployment point. A greedy insertion adds the site and the position with the largest gain of V until no insertion raises V. A local search then draws 200 moves from five kinds, namely reversal of a segment, relocation of a visit, exchange of a visited site with an open one, removal of one to three visits followed by reinsertion as in large neighborhood search [35], and insertion of one more site. It keeps a move that raises V or that keeps V and lowers U. Program (10) then refines the waypoints and durations of the best order. When the refinement frees part of the budget, a second insertion pass uses it. The search over V can stop at a local optimum. TACO therefore runs two more searches of the same form. One accepts only orders with $U \leq 1$. The other accepts only orders whose mean time and energy fit both budgets with a 10% margin. A single-station plan is one more candidate. While the ROV is on deck, its station is the point of a 15 m grid whose greedy tour within reach has the largest value. After deployment it holds the catamaran at its current position. Program (10) refines this candidate as well. TACO returns the candidate with the largest V. Its plan therefore has at least the V of the single-station plan. Algorithm 1 summarizes one planning step.

During execution both vehicles track reference points that move along their legs with one common progress over the planned duration t_k. After every sample TACO plans again from the measured time, the measured energy and the open sites. This receding-horizon scheme follows planners for uncertain environments [36], [37]. The planner runs in a process of its own. The control loop therefore never waits for it. While it plans, the ROV and the catamaran hold their positions. This

Algorithm 1 One planning step of TACO

Input: open sites, positions of both vehicles, used time T_0 and energy E_0
Output: order $\boldsymbol{\pi}$, waypoints $\mathbf{B}$, durations $\mathbf{t}$

for each acceptance rule in $\{V$ rises, $U \leq 1$, 10% margin$\}$ **do**
  $\boldsymbol{\pi} \leftarrow$ greedy insertion with projected waypoints under the rule
  for 200 moves **do**
    draw a move, form $\boldsymbol{\pi}'$ and project its waypoints onto the discs (6)
    cut $\boldsymbol{\pi}'$ before its first visit with $q_k < 0.5$ in (8)
    if the rule accepts $\boldsymbol{\pi}'$ over $\boldsymbol{\pi}$ **then** $\boldsymbol{\pi} \leftarrow \boldsymbol{\pi}'$
  end for
  $(\mathbf{B}, \mathbf{t}) \leftarrow$ solve (10) for $\boldsymbol{\pi}$
  insert further sites under the rule and solve (10) again
end for
add the single-station plan and its solution of (10)
return the candidate $(\boldsymbol{\pi}, \mathbf{B}, \mathbf{t})$ with the largest V

time counts against the budget. A return guard runs below every planner. The ROV stops sampling and returns when γ times the time or energy of the ascent from its current point no longer fits the budget.

V. Experiments

A. Experimental Setup

Fig. 5 shows the three scenarios. Scenarios 1 and 2 test the closed loop at a reduced scale with flight-controller hardware. Scenario 3 tests TACO at the scale of a work-class ROV and its mother ship. Scenarios 1 and 2 run in the six-degree-of-freedom native core of NezhaSim2. The core integrates each vehicle with a fourth-order Runge-Kutta scheme that includes the added mass at a step of 4 ms. The ROV of Scenarios 1 and 2 is the BlueROV2 Heavy reference of the simulator. It has a mass of 13.5 kg, eight T200 thrusters on curves fitted to the manufacturer table and damping identified in a test tank [22]. Its thrust power follows the manufacturer table at 20 V as $1.163|F|^{1.5}$ W forward and $1.775|F|^{1.5}$ W reverse with F in newtons. The two fits have coefficients of determination of 0.998 and 0.999. The catamaran is the surface vessel of the simulator scaled by a length ratio of three under Froude similarity, to a 2.7 m hull of 324 kg with two 300 N stern drives. A lumped-mass tether of 70 m with 24 segments, a diameter of 7.6 mm and a normal drag coefficient of 1.2 links the two vehicles and returns its end forces to both plants [21], [22]. The ROV draws its hotel load of 60 W and a sampling tool of 150 W through a tether efficiency of 0.92. The catamaran adds a hotel load of 80 W and the actuator-disc power of its drives.

The test field of Scenarios 1 and 2 is a 240 m by 160 m window of the multibeam heightfield of one dive on a crust seamount. We shifted it to a mean depth of 35 m and kept its relief of 14.5 m. The current is the profile measured by the ship-mounted acoustic Doppler profiler during that dive, 0.15 to 0.19 m/s toward the west in its upper bins. Scenario 1 uses this profile with waves of 0.2 m significant height. Scenario 2 doubles the current, raises the waves to 0.5 m and draws the sampling durations from the lognormal shape fitted to the field stations of Section V-E. Sampling lasts 45 s on average, with a coefficient of variation of 0.15 in Scenario 1 and 0.99 in Scenario 2. Scenario 1 has 16 sites and Scenario 2 has 20 sites, with a spacing of at least 18 m and values from 1 to 10 drawn from a smooth grade field. Both scenarios have a time budget of 900 s, with energy budgets of 600 kJ and 650 kJ. The catamaran starts 140 m west of the field center with the ROV on deck.

We fitted the leg energy model (7) to 37 legs of seven calibration missions in Scenario 1 and to 33 legs of ten missions in Scenario 2. These missions ran on three instances that are not used for evaluation. The fit had coefficients of determination of 0.93 and 0.91 for the total leg energy. Its tether-load coefficient a_3 was 6697 and 2367 J s/m^2. A catamaran move of 40 m in 100 s therefore cost the ROV about 107 kJ in Scenario 1. The leg deviations had coefficients of variation c_e of 0.35 and 0.48 for energy and c_t of 0.05 and 0.12 for duration. TACO uses $\epsilon = 0.05$ in (9).

Hardware-in-the-loop tests have checked planners of surface vehicles [37] and the computing of underwater vehicles [29], [38] before field trials. In our runs a Pixhawk 6C with PX4 v1.17 [39] receives simulated inertial, magnetic and pressure readings at 250 Hz. It also receives a position fix at 10 Hz. The fix stands for the navigation solution of an ultra-short baseline and Doppler log, with a noise of 0.15 m. Depth reaches the board through its barometer channel as an equivalent altitude. PX4 holds the attitude and allocates a three-axis thrust demand to the eight reversible thrusters with the geometry of the ROV. The guidance law on the companion computer sends that demand with the heading setpoint at 50 Hz. The board reboots before every run. Each scenario has eight instances in hardware-in-the-loop. For the ablation and for a larger sample, we also ran 16 instances per scenario with the Python replica of the PX4 loops in place of the board. In these replica runs the vehicles hold for the processor time of each planning call. The simulator, planners and replica ran on one laptop processor. We rendered logged states of the replica runs in Isaac Sim, with the tether drawn through the logged nodes of its lumped-mass model. The renderings draw the catamaran as the mother ship of Fig. 3 at its scale of 2.7 m.

We compare TACO with two baselines that use the same cost model, executor and return guard. The fixed-base planner carries the ROV on deck to one station and holds it for the whole dive, as with a station-keeping catamaran [19]. It picks the station on a 15 m grid by the value of a greedy tour within reach and then runs the insertion and local search of TACO with fixed waypoints. The cluster-route planner follows the cluster-first route-second scheme [33], [34]. It covers the sites with reach discs centered on the same grid. Nearest neighbor and 2-opt order the stations and the sites inside each station. The planner then drops the site with the smallest value per freed utilization until the plan fits. Both baselines accept a plan whose mean time and energy fit with a 10% margin. Like TACO, they replan after every sample. We report the value kept, the value of the samples of a mission recovered within both budgets. We also report the share of missions within both budgets with the time and energy used.

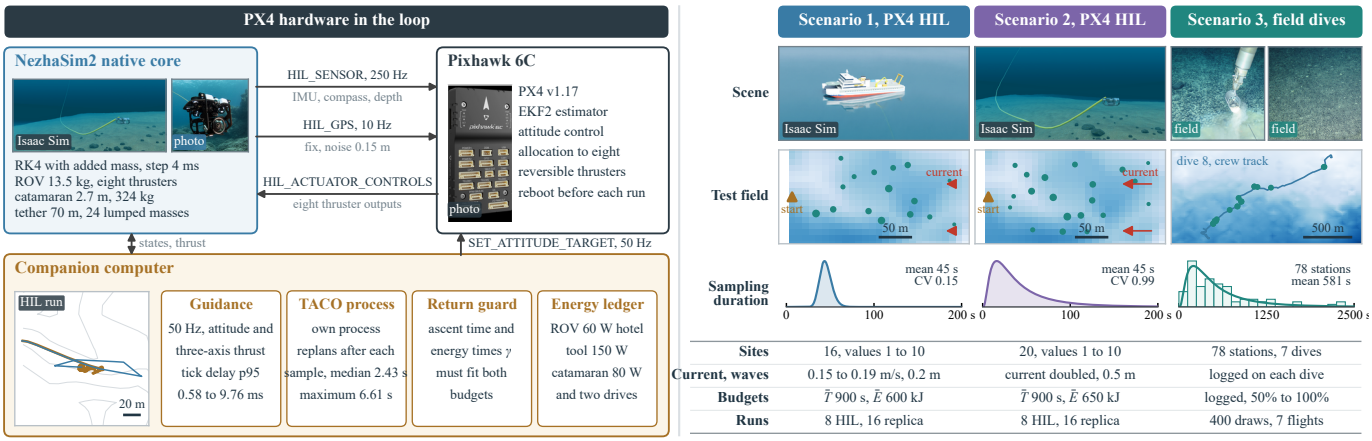


Fig. 5. Experiment design. In hardware in the loop the Pixhawk 6C runs PX4 against the native core of NezhaSim2 over the MAVLink messages shown with their rates. The companion computer runs the guidance law, TACO, the return guard and the energy ledger. Its inset shows the paths of a hardware-in-the-loop run on instance 1 of Scenario 1. The columns on the right give the scene, the test field, the sampling durations and the settings of each scenario. The renderings show logged states of replica runs with the catamaran drawn as the mother ship at its scale of 2.7 m. The map of Scenario 3 shows the logged track and the sampling stations of one field dive. The photographs of Scenario 3 come from a field dive. The photograph of a BlueROV2 is by Yoleeth under the CC BY-SA 4.0 license. The image of the Pixhawk 6C comes from the PX4 User Guide under the CC BY 4.0 license.

TABLE II
HARDWARE-IN-THE-LOOP RUNS OVER EIGHT INSTANCES PER SCENARIO

	Scenario 1				Scenario 2			
Planner	Kept	Within	E (kJ)	Base (m)	Kept	Within	E (kJ)	Base (m)
TACO	**33.1**	**8/8**	**458**	**126**	**29.0**	**8/8**	**479**	**180**
Fixed base	31.2	8/8	431	127	24.8	8/8	510	156
Cluster-route	19.2	8/8	312	157	20.2	8/8	418	185

B. Scenario 1: Low Sampling Spread

Table II lists the hardware-in-the-loop runs. Its column Kept gives the mean value kept per instance. The column Within counts the runs that ended within both budgets. The columns E and Base give the mean energy used and the mean distance moved by the catamaran. All 24 runs of Scenario 1 ended within both budgets. TACO kept 33.1 per instance against 31.2 for the fixed base and 19.2 for the cluster-route planner. Its value kept was 6.2% above that of the fixed base. TACO beat the fixed base on six instances and tied it on two, as shown in Fig. 6(b). On average a dive of TACO kept 4.5 samples with 89.9% of the time budget and 76.3% of the energy budget. The cluster-route planner kept 2.4 samples per dive. Its dives left 36% of the time budget unused.

Fig. 6 shows the runs of both scenarios. On instance 1 both TACO and the fixed base kept four samples, as shown in Fig. 6(a). The fixed base carried the ROV 114 m on deck to its station near the center of the field. TACO deployed the ROV after a deck transfer of 94 m. It then moved the catamaran 20 m west with the ROV in the water. With this move the ROV reached the site of value 10 at the western end of the field. The fixed base took two northern sites of 9.4 and 4.5 instead. The cluster-route planner kept less than TACO and the fixed base on every instance.

C. Scenario 2: Field Sampling Spread

All 24 runs of Scenario 2 also ended within both budgets. TACO kept 29.0 per instance against 24.8 for the fixed base and 20.2 for the cluster-route planner. This was 17.0% more than the fixed base. TACO beat the fixed base on five instances, tied it on one and lost on two. It beat the cluster-route planner on all eight instances, as shown in Fig. 6(d). Fig. 6(c) shows how each run spent the two budgets. On average a dive of TACO used 82.0% of the time budget and 73.7% of the energy budget for 3.6 samples. The fixed base used 76.7% of the time budget and 78.5% of the energy budget for 3.1 samples. On instance 8 the return guard ended one dive of each baseline.

D. Mission Traces in Hardware in the Loop

Figs. 7(a) and 7(b) follow TACO on instance 4 of Scenario 2. It planned four times. Its three samples had a value of 25.8. Sampled once per second, its tether tension peaked at 33.3 N in a transit. Every predicted end point fell inside both budgets. The recovery came within the spread of the last prediction. Over the 48 runs the remaining time predicted by a plan had a median absolute error of 5.2% in Scenario 1 and 12.7% in Scenario 2, as shown in Fig. 7(d). The remaining energy was predicted 8.6% and 25.6% too high at the median. The plans therefore left part of the energy budget unused, as in Fig. 6(c). The tether loss took 5.5% to 6.1% of the energy of every planner, as shown in Fig. 7(c). The catamaran took 23% to 32%.

In hardware in the loop each replan of TACO took a median of 2.43 s of wall time with a maximum of 6.61 s. The 95th percentile of the tick delay of the 50 Hz guidance loop was 0.58 to 9.76 ms per run. The replica reproduced the value kept of 47 of the 48 runs. On these runs the median differences were 0.10% in time and 0.66% in energy. The exception was the fixed base on instance 1 of Scenario 2. In hardware in the loop it kept one more sample than in the replica.

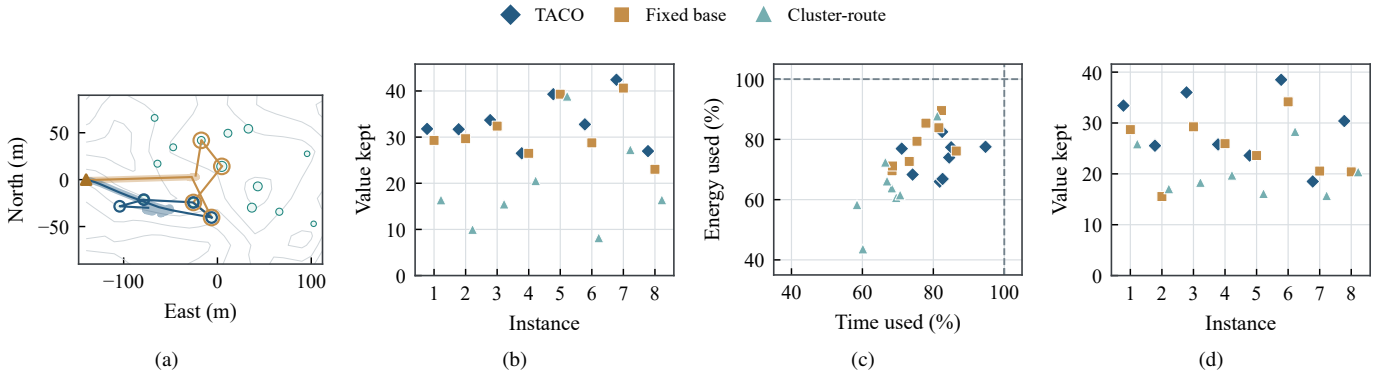


Fig. 6. Hardware-in-the-loop runs. (a) Paths on instance 1 of Scenario 1. Thick lines show the catamaran. Thin lines show the ROV. Rings mark the samples kept. (b) Value kept on each instance of Scenario 1. (c) Time and energy used by each run of Scenario 2 as shares of the budgets. (d) Value kept on each instance of Scenario 2.

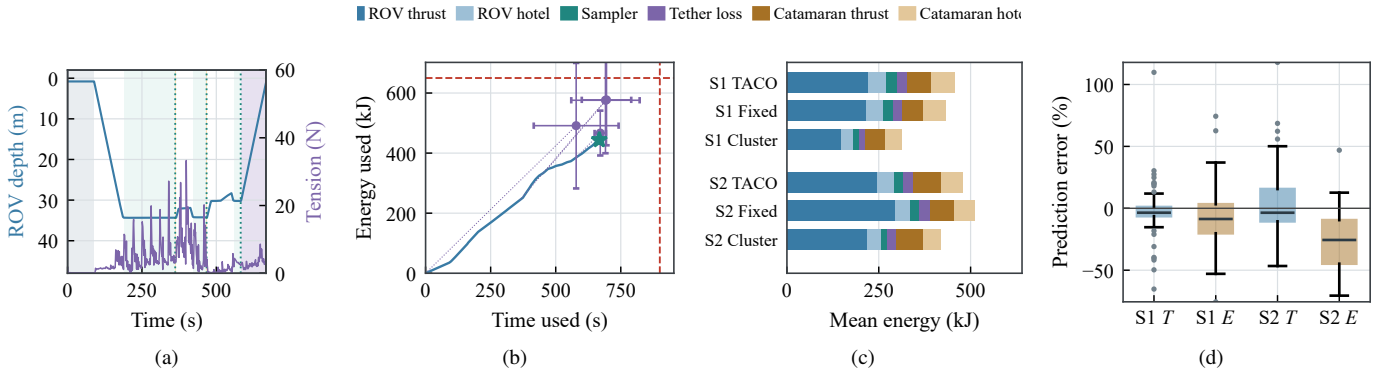


Fig. 7. Hardware-in-the-loop mission traces. (a) Depth of the ROV and tether tension on instance 4 of Scenario 2 flown by TACO. Shading marks the deck, planning, sampling and ascent phases. Dotted lines mark the samples. (b) Time and energy used on the same run. Dashed lines are the budgets. Circles are the end points predicted by its four plans with two standard deviations. The star is the recovery. (c) Mean energy of each planner by component over the eight runs per scenario. (d) Error of the remaining time T and energy E predicted at every replan of the 48 runs.

E. Scenario 3: Field Dives of a Work-Class ROV

Scenario 3 asks whether TACO is feasible at the scale of a work-class ROV and how it compares with the crew of real dives. We read the main-control logs of seven dives of a work-class ROV on two seamounts at depths of 1 to 3 km. The logs hold the thruster commands, the pressures of the main and auxiliary hydraulic pumps, every valve of the manipulator and tool circuits, the depth, the altitude and the ship position at about 9 Hz. We count a landed station with at least 2 s of manipulator or tool valve activity as a sampling station. The field sheets give the times of nine ore samples, namely crusts, nodules and crust-covered basalt. Eight of them fall inside detected stations. All eight meet this rule. The ninth falls in an interval without altitude lock, where no station was detected. Fig. 8 shows the bottom phase of one dive with the photographs of its still camera. Of the five photographs of manipulator or landed work that we hold from this dive, four fall inside detected sampling stations. The fifth is the scoop sample of Fig. 8, taken 90 s after the end of a station. The logs record no motor current. We therefore estimate the hydraulic power as pump pressure times delivered flow. The main flow is the rated flow of the main pump scaled by the summed thruster command. The auxiliary flow is the rated flow of the auxiliary pump scaled by the share of open functions.

Fig. 9 and Table III give the results. The 78 sampling stations lasted 581 s on average with a median of 415 s and a standard deviation of 504 s. A lognormal distribution with a shape of 0.83 and a median of 416 s described them, as shown in Fig. 9(a). TACO uses its mean and spread for every site. The 95th percentile of the 20,155 ship-to-ROV offsets during the bottom phases was 321 m, as shown in Fig. 9(b). TACO takes this value as the reach. The other terms came from the operating modes of the logs. The ROV crossed between stations at 0.167 m/s with an overhead of 147 s per leg. It drew 31.1 kW in transit and 42.1 kW while it sampled. It descended at 0.41 m/s and ascended at 0.46 m/s. On every held-out dive, a model fitted to the other six dives predicted the dive duration and energy along the real station sequence. The median absolute errors were 8.2% for duration and 8.5% for energy, as shown in Fig. 9(c). The largest errors of 23.5% and 30.5% came from the dive with only four sampling stations in 8.9 h.

Each planner then planned every dive over its own sampling stations from the ship position at launch. The budgets were the logged dive duration and estimated energy. A second case cut both to 70%. The catamaran stands for the support ship and moves at up to 0.5 m/s. The energy budget covers the hydraulic energy of the ROV. We drew 400 executions of every plan with lognormal sampling durations and leg deviations. The leg deviations had a coefficient of variation of 0.38 for

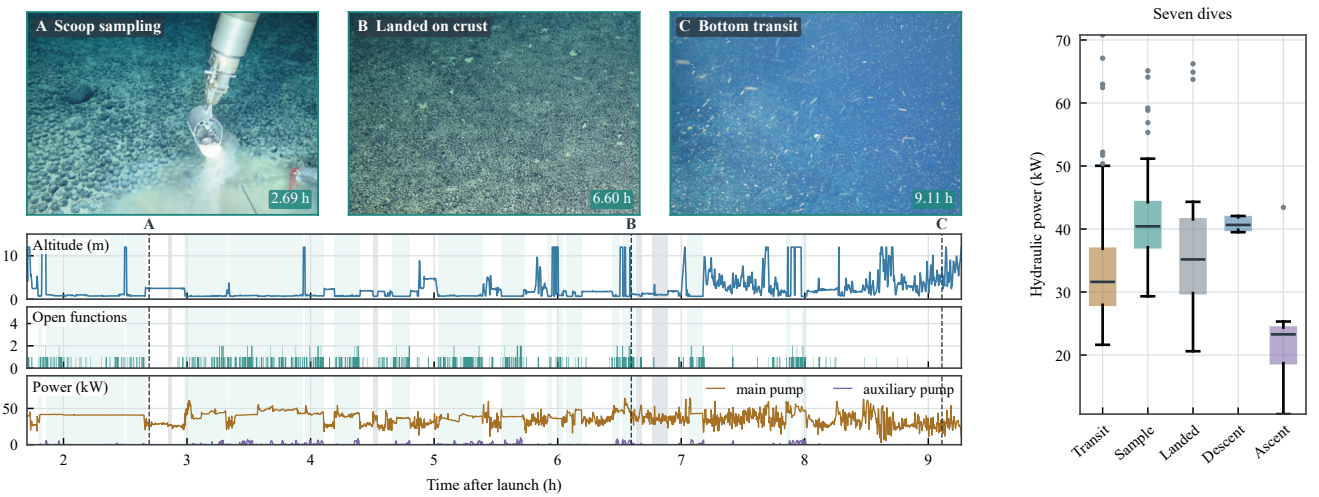


Fig. 8. Bottom phase of a field dive of the work-class ROV. The photographs come from its still camera with their times after launch. The traces below give the altitude, the open manipulator and tool functions and the estimated hydraulic power. Green bands mark detected sampling stations. Grey bands mark other landed stations. Dashed lines mark the photographs. The right panel gives the hydraulic power of every leg and station of the seven dives by operating mode. The maker mark on the manipulator is blurred.

duration, which is the speed spread of 38 direct transits in the logs. Their coefficient for energy was 0.44. The return guard of Section IV-C ran in every draw. With the logged budgets TACO kept 70.6 samples over the seven dives, 25% more than the fixed base. The cluster-route planner kept 70.3 samples. Counting a station only when the ascent after it still fit both budgets, the crew kept 76 of its 78 sampling stations. At 70% of the budgets TACO kept 47.1 samples. This was 7.0% more than the fixed base and 17% more than the cluster-route planner. The crew had finished 45 sampling stations by the time it had used 70% of its budgets less the ascent. TACO took more risk at this budget. Of its draws 91.1% ended within both budgets, against 92.6% and 94.1% for the baselines. TACO planned a dive in a median of 3.4 s with the logged budgets and 6.3 s at 70% of them.

We then flew the plans of TACO for all seven dives at 70% of their budgets with the identified model of this ROV in NezhaSim2. The model flew over the multibeam terrain of each dive with seabed contact, in the measured current and density. Its task list held a descent, terrain-following transits, landed holds of 581 s with push-down and the ascent. All seven dives completed their 4 to 14 visits within both budgets. They took 4.4% to 12.0% less time than planned. Most of this difference is the leg overhead of 147 s fitted to the legs of the crew, which the automatic legs did not need. Their estimated energy differed from the plan by -13.3% to 13.1%, with a median of 4.9%. The closest dive used 99.7% of its energy budget, as shown in Fig. 9(g). On the flight of Fig. 9(e) the model kept its center at a median of 3.0 m above the seabed in transit and 1.5 m in the landed holds, as shown in Fig. 9(f). Fig. 9(h) splits the dive time by mode. The crew spent a median of 46.0% of its dive time in transit between stations against 14.4% for the flights of TACO. The transits of the crew include survey lines. The landed shares were 30.6% and 32.3%. Descent and ascent took medians of 28.6% and 24.8% of the shorter flights against 16.2% and 11.4% of the logged dives.

TABLE III
SCENARIO 3 OVER SEVEN FIELD DIVES

	Logged budgets			70% of the budgets		
Planner	Kept	Within (%)	Base (km)	Kept	Within (%)	Base (km)
TACO	**70.6**	**97.4**	**0.83**	**47.1**	**91.1**	**0.53**
Fixed base	56.6	99.4	0.52	44.0	92.6	0.48
Cluster-route	70.3	96.1	1.02	40.2	94.1	0.63
Crew, logged	76		3.17	45		

F. Ablation

We ran four variants of TACO on the 16 replica instances of each scenario. The margin variant runs only the search that accepts orders whose mean time and energy fit with a 10% margin. The chance variant runs only the search that accepts orders with $U \leq 1$. Neither of the two adds the single-station candidate. The projected variant keeps the projected waypoints of the order search and skips program (10). The open-loop variant follows its first plan for the whole dive. Table IV lists the value kept by these variants and by the two baselines on the same instances. Its column Δ gives the mean paired difference from TACO. The column L/T/W counts the instances on which a method lost to TACO, tied with it or beat it. Fig. 10 shows every paired difference with the 95% interval of the mean. All 224 replica runs ended within both budgets.

TACO kept the most value in both scenarios. Its gains over the fixed base were 6.1% and 12.2%, with intervals that excluded zero. Against the cluster-route planner TACO gained 50.4% and 43.3%.

Each single search kept less than the full order search. With the wide spread of sampling durations in Scenario 2, the bound $U \leq 1$ admitted few visits. There the dives of the chance variant used 65.7% of the time budget and kept 2.56

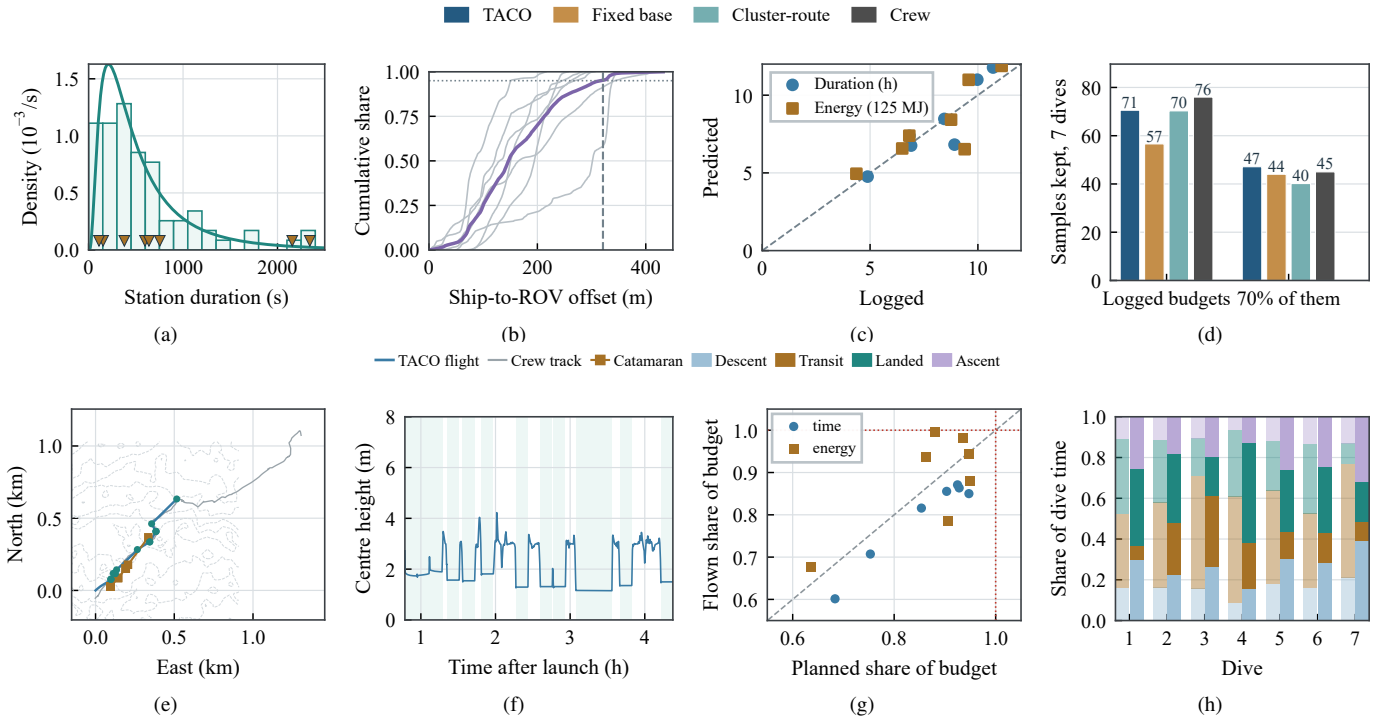


Fig. 9. Scenario 3 on the field dives. (a) Durations of the 78 sampling stations with the fitted lognormal. Triangles mark the stations of logged ore samples. (b) Ship-to-ROV offset during the bottom phases. Grey lines are single dives. The purple line pools them. The dashed line marks the 95th percentile. (c) Predicted against logged duration and energy of held-out dives, with energy in units of 125 MJ. (d) Samples kept over the seven dives. (e) Path flown by TACO in NezhaSim2 on one dive at 70% of its budgets and the USBL track of the crew in its bottom phase, relative to the launch point. Dots are the sampling stations. (f) Height of the vehicle center above the seabed on that flight. Bands mark the landed holds. (g) Planned against flown time and energy of the seven flights as shares of their budgets. (h) Shares of the dive time for the crew in the lighter left bars and for the flights of TACO in the right bars.

TABLE IV
REPLICA RUNS OVER 16 INSTANCES PER SCENARIO

	Scenario 1			Scenario 2		
Method	Kept	Δ	L/T/W	Kept	Δ	L/T/W
TACO	**32.18**			**27.81**		
Margin only	31.81	−0.37	7/7/2	26.22	−1.58	4/9/3
Chance only	31.12	−1.06	4/11/1	21.84	−5.97	12/3/1
Projected waypoints	32.18	0.00	0/16/0	25.79	−2.01	5/8/3
No replanning	31.81	−0.37	3/13/0	26.50	−1.31	6/7/3
Fixed base	30.34	−1.84	11/5/0	24.78	−3.03	8/5/3
Cluster-route	21.39	−10.79	13/3/0	19.40	−8.40	16/0/0

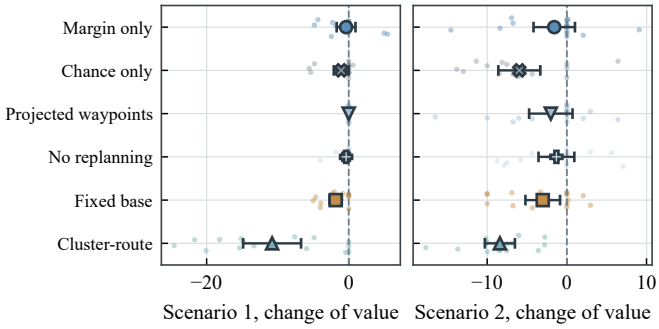


Fig. 10. Change of the value kept against TACO on the replica instances. Dots are paired instances. Markers give the mean paired difference with its 95% interval.

samples on average. TACO used 80.1% of the time budget and kept 3.62 samples. Selection by V over the candidates of both searches thus kept more value than either search alone.

Program (10) mattered in Scenario 2. The projected variant tied TACO on all 16 instances of Scenario 1. In Scenario 2 it lost 2.01 with 3.31 samples per dive. Replanning also gained more in Scenario 2. The open-loop variant lost 0.37 in Scenario 1, where it beat TACO on no instance. In Scenario 2 it lost 1.31, with six losses and three wins. The intervals of the margin, projected and open-loop variants included zero in both scenarios.

G. Optimality, Calibration and Budget Level

We drew 20 instances of eight sites from each PX4 scenario with the budgets of that scenario. An exhaustive search scored every order of one to eight visits with projected waypoints under the rule $q_k \geq 0.5$. Program (10) then refined the 200 orders with the largest V. Fig. 11(c) compares the first plan of TACO and of the fixed base with the best plan found. TACO matched it within 0.05% on 14 and 15 of the 20 instances. On seven instances TACO exceeded the order search. On five of them it chose the same order with better waypoints. On the other two it chose an order that the projected waypoints had ruled out. Its mean shortfall was 1.3% in Scenario 1 and 0.5% in Scenario 2, with a largest shortfall of 11.3%. The fixed base fell short by 7.0% and 8.5% on average. The exhaustive search took 60 s and 35 s per instance against 4.1 s and 3.3 s for TACO. Fig. 11(d) gives the time of the first plan of TACO on

fields of Scenario 1 with more sites. It took a median of 4.1 s for 10 sites, 28.6 s for 40 sites and 69.1 s for 80 sites.

In the budget sweep below, TACO planned 342 visits of the field dives. Fig. 11(b) compares their predicted q_k with the share of the 400 draws in which each visit counted. The shares rose with q_k but fell 3 to 8 points below it. The lognormal sampling durations of the field have a heavier right tail than the normal sums behind (8).

We varied the budgets of Scenario 3 from 50% to 100% of the logged ones, as shown in Fig. 11(a). TACO kept the most samples of the three planners at every level, from 21.5 at 50% to 70.6 at 100%. Its gain over the fixed base grew from 9.5% at half the budgets to 24.6% at the full budgets. A longer dive reaches more stations beyond one reach disc. The gain over the cluster-route planner peaked at 17.4% at 70% of the budgets. The crew kept fewer samples than TACO up to 70% of the budgets and more above it, with 15 samples at 50% and 76 at 100%. At 80% and 90% the margin variant of TACO kept 0.3 and 0.2 more samples than TACO.

VI. Discussion

A. Where the Coupling Pays Off

The gain of TACO grew with the ratio of the site spread to the tether reach. On the test field of Scenarios 1 and 2, one reach disc of about 50 m covered most of the high-value sites of an instance. The fixed base therefore came close to TACO. In hardware in the loop the gain of TACO over the fixed base rose from 6.2% in Scenario 1 to 17.0% in Scenario 2. Scenario 2 spreads the sampling durations as in the field. On the field dives the sites spread over kilometers against a reach of 321 m. There the fixed base kept 20% fewer samples than TACO with the logged budgets. In the budget sweep the gain rose with the budget, from 9.5% to 24.6%. The tether load sets how far a coupled plan should move the base. Because a move of 40 m in 100 s cost the ROV about 107 kJ in Scenario 1, TACO moved the catamaran mostly with the ROV still on deck. In hardware in the loop the catamaran of TACO moved 94 m and 152 m with the ROV on deck in the two scenarios. With the ROV in the water it moved 32 m and 28 m. The convex program still moved the catamaran when a shorter descent or ascent of the ROV saved more than the tether load cost, as in Fig. 4. The risk-weighted value mattered when the sampling durations spread widely. In the replica runs of Scenario 2 the chance variant kept 21.84 against 27.81 for TACO. Its whole-plan constraint protects the last visit of a plan. The return guard and the next replanning step already drop that visit when it no longer fits.

B. Limitations

Five limits bound these results. First, the tether load enters the planner as a fitted term of the leg energy. The planner does not simulate the cable. The fit explained 91% to 93% of the leg energy. Second, the hydraulic power of the work-class ROV is an estimate from pump pressures and valve commands. Its logs hold no motor current. The relative cost of transit, sampling and vertical travel is therefore better supported than its absolute level. Third, Scenario 3 compares the planners by Monte Carlo draws of fitted durations and deviations. Its closed-loop flights covered the plans of TACO only. They left out the support ship and its umbilical. Fourth, the hardware-in-the-loop runs replace the navigation sensors of the ROV by simulated fixes. They also use the multicopter firmware of PX4 with the thruster geometry of the ROV. The test field keeps the relief of a deep seamount at a depth of 35 m with a catamaran of 2.7 m. Fifth, the completion probabilities of TACO were 3 to 8 points too high on the field dives. A duration model with the heavier tail of the field stations could reduce this bias.

VII. Conclusion

We have proposed the Tether-Aware Coupled Orienteering (TACO) planner for a catamaran base that deploys a tethered ROV to sample the seabed under a time budget and an energy budget drawn from one store. TACO has turned the tether constraint into one reach disc per visit, charged the load of a moving tether to the leg energy and placed the catamaran waypoints with the leg durations by a convex program. It has selected the visits by their value weighted with the probability of completing them within both budgets and has replanned after every sample. In PX4 hardware-in-the-loop tests, TACO has kept 6.2% and 17.0% more sample value than a fixed catamaran base. All 48 hardware-in-the-loop runs have ended within both budgets. On seven field dives of a work-class ROV with budgets cut to 70%, TACO has kept 47.1 samples against 44.0 for the fixed base, 40.2 for the cluster-route planner and 45 for the crew. All seven of its plans flown in closed loop have ended within both budgets. On 29 of 40 small instances TACO has matched or exceeded an exhaustive search over visit orders.

In our future work, we will test the tether-load model in sea trials with a small catamaran and a tethered ROV. We will also update the sampling-duration distribution of a site from the samples taken during the dive.

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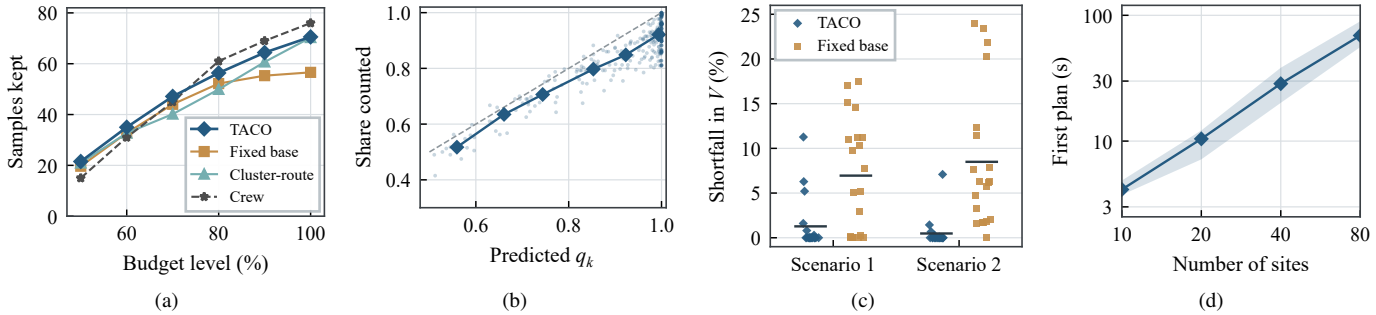


Fig. 11. Budget level, calibration and optimality. (a) Samples kept over the seven field dives against the budget level. (b) Share of 400 draws in which a visit planned by TACO counted against its predicted probability q_k. The line joins the bin means. (c) Shortfall of the risk-weighted value of the first plan from the best plan found on 40 instances of eight sites. Bars are means. (d) Time of the first plan of TACO against the number of sites, with the median and the range of five fields.

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